

GOMEZYANI OBED SILWEYA

2303454623

RESEARCH TOPIC: STUDY OF INDIGENOUS BUILDING MATERIALS AND PRACTICES ACROSS ZAMBIAS PROVINCES.

STATEMENT OF THE PROBLEM

In as much as urbanization and modernism is growing Fastly, the diverse range of locally building materials like stone, earth, clay, timber, lime, reeds, and grass is suddenly declining, which if looked back historically they were used in different communities and tribes according to their local environment, climate and cultural practices.

Assuming the cause of the decline is because indigenous construction represents poverty or outdated practices, of which this objective is void due to inadequate documentation of traditional knowledge. This decline has resulted in loss of value in the practice, less opportunities to use locally sourced materials in contemporary architecture.

The issue has now increased the dependency of manufactured materials, which costs a lot in the construction sector, not only that it raises concerns on environmental health impacts, while silencing the identity of local architecture. Henceforth, it calls a huge need to examine the availability and traditional utilization of indigenous building materials across Zambia, and look on how they can be adapted to contemporary construction.

This sturdy will examine how combining both modern building technology and traditional knowledge can promote sustainability, cultural identity, affordability, and natural aesthetics in addition to giving locally available materials a meaningful signature in Zambia's architectural future development.

SCOPE OF THE STUDY

The study will focus on the indigenous building materials and their practices across the ten provinces of zambia, thus targeting different types of indigenous materials that where used long ago and how effective they can help us in the Morden development in conjunction with minimizing industrial building materials. This study will involve the support and emphasis from personnel in the industry like architects, engineers, contractors, and crafts men.

Focusing on all ten provinces in zambia and each province having its own common unique indigenous building material. This will be evoked by first specifying each material, and understanding how the specie reacts in that geographic area, additionally

how they can be enhanced to reach the required standard of strength. The study will also dive in on how preserving measures can be obtained and how efficient it can be to the construction sector; it will also draw knowledge from local communities and construction professionals.

This research will approximately take seven-eight weeks to be done covering all necessary knowledge and finding that indigenous plants outshine. However, limitations of some evidence and actual species will be encountered due to some being found in areas for animal reserves.

REFERENCE

Vellinga, M., Oliver, P. and Bridge, A., 2024. *Atlas of vernacular architecture of the world*. Taylor & Francis.

Papanek, V., 2022. *The green imperative: ecology and ethics in design and architecture*. Thames & Hudson.

Yanou, M.P., Ros-Tonen, M.A., Reed, J., Nakwenda, S. and Sunderland, T., 2024. The hybridisation, resilience, and loss of local knowledge and natural resource management in Zambia. *Human Ecology*, 52(5), pp.1087-1105.

Klemczak, B., Kucharczyk-Brus, B., Sulimowska, A. and Radziejewicz-Winnicki, R., 2024. Historical evolution and current developments in building thermal insulation materials—A review. *Energies*, 17(22), p.5535.

Huang, J., Lucash, M.S., Scheller, R.M. and Klippel, A., 2021. Walking through the forests of the future: using data-driven virtual reality to visualize forests under climate change. *International Journal of Geographical Information Science*, 35(6), pp.1155-1178.

Ngowi, A. and Awuzie, B.O., 2026. Unveiling the significance of indigenous knowledge in shaping sustainable construction practices. In *Research Companion to the Construction Industry in Developing Countries* (pp. 248-269). Edward Elgar Publishing.